Additional file 1: Supplementary tables and figures for "HANFIS: A Hierarchical Adaptive Neuro-Fuzzy Inference System for Evidence-Based Dynamic Cybersecurity Risk Scoring in Cloud-Native Kubernetes Deployments"

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Abstract

Supplementary tables and figures moved out of the main text to shorten it. Each item keeps its original caption and values.

Supplementary tables

Supplementary figures

Supplementary notes

Notes S1 to S7 reproduce the original wording of the corresponding main-text sections as they stood on 3 November 2026. The main text has since been revised; for example,

Table S1 Experimental preparation, acceptance criteria, and methodological purpose.

Preparation element	Acceptance criterion	Methodological purpose	Evidence (Runs 7–16)
Cluster health	Two Kubernetes nodes in Ready state	Excludes measurements obtained during node degradation or loss of schedulability	Gate passed in all ten pairs
Baseline workload P0	Two of two replicas Ready	Holds application capacity constant across monitoring conditions	Gate passed in all ten pairs
Service health	Internal HTTP response code 200	Confirms that the measurement target remains operational before and after state transitions	Gate passed in all ten pairs
Monitoring ON state	Five of five pods Running and fully Ready; Prometheus 1/1 Ready	Defines a complete observability treatment rather than a partially deployed stack	Gate passed in all ten pairs
Monitoring OFF state	Zero monitoring pods remaining	Defines a valid control condition without residual collector workloads	Gate passed in all ten pairs
Protocol freeze	Scripts, controller definitions, services, system fingerprint, and replication plan present	Prevents unrecorded procedural drift between runs	Applied to Runs 7–16; two aborted attempts logged as incidents
Integrity control	SHA-256 manifests validate without mismatch	Detects post hoc alteration or corruption of experimental artefacts	SHA-256 manifest retained for each of the ten run archives
Counterbalancing	Prespecified ON–OFF or OFF–ON order	Reduces order, warm-up, cache, and time-dependent bias	Five ON–OFF and five OFF–ON pairs

the comparison with FlatANFISMatched that these notes label H2 is labelled H2/H3 in the main text.

Supplementary note S1: Experimental platform and telemetry validation (original text of Section 3.3)

The experimental workflow was implemented in Ubuntu 24.04 under Windows Subsystem for Linux 2 (WSL 2), with Visual Studio Code connected directly to the WSL environment. Docker Desktop provided the container runtime and Kubernetes integration. The research workspace was maintained under a dedicated project hierarchy, while shell procedures were stored as versionable `.sh` files rather than being executed only as transient terminal commands. This decision improved procedural transparency,

Table S2 E1 ablation: membership family × monotonicity, 5 seeds per cell.

Scenario	MF family	Monotone	Params.	RMSE (mean ± SD)	R^2 (mean)
S1 (smooth)	Gauss	no	176	0.093 ± 0.021	1.000
S1 (smooth)	Gauss	yes	176	0.411 ± 0.246	0.999
S1 (smooth)	Ruspini	no	147	0.043 ± 0.052	1.000
S1 (smooth)	Ruspini	yes	147	0.073 ± 0.066	1.000
S2 (threshold)	Gauss	no	176	0.117 ± 0.034	1.000
S2 (threshold)	Gauss	yes	176	6.067 ± 1.524	0.879
S2 (threshold)	Ruspini	no	147	0.646 ± 0.051	0.999
S2 (threshold)	Ruspini	yes	147	2.126 ± 0.906	0.983

Table S3 Post hoc sensitivity of the confirmatory comparison to the EPSS model change: full run versus weeks whose labels all pre-date 2026-06-15 (skill score, 95% group-clustered bootstrap CI).

Hypothesis	Comparator	Full run	Pre-change weeks
H1a	Single CVSS score	0.967 [0.951, 0.980]	0.973 [0.939, 0.991]
H1b	Persistence (4-week)	0.015 [−0.607, 0.246]	−0.055 [−2.731, 0.201]
H1c	Ridge, 7 inputs	0.696 [0.527, 0.816]	0.728 [0.484, 0.901]
H2/H3	FlatANFISMatched	0.665 [0.269, 0.844]	0.913 [0.753, 0.974]
H2′	Flat ANFIS (full)	0.253	0.441

Table S4 Explanation-fidelity checks on the 25 saved models. Pairs are the positive-control images P0-IF with and without the active filter; S^+ denotes the 23 weeks in which exploitability differs between the two.

Check	Criterion	Result	Outcome
H6-0 trace vs. inference	$\lvert\Delta\rvert \le 10^{-9}$, all test rows	0	pass
H6-neg identical inputs	$\lvert\Delta\rvert \le 10^{-9}$, 34 weeks × 25 models	0	pass
H6a direction (monotone)	$\hat{y}_{\text{on}} - \hat{y}_{\text{off}} \ge -10^{-9}$ on S^+	minimum 0	pass
H6b panel structure	panel E unchanged, panel V changes	0 and 2.9×10^{-4}	pass
H6c deletion test	ratio ≥ 2 and proportion ≥ 0.90	4.11 and 0.835	fail

reduced copy-and-paste errors, and allowed every experimental gate to be re-examined without reconstructing the original command sequence.

The Kubernetes testbed comprised two Ready nodes and a baseline application, denoted P0, deployed with two ready replicas in the `hanfis-research` namespace. A load-generator pod accessed P0 through an internal Kubernetes service. Service availability was verified using an HTTP request from inside the cluster; status code 200 was required at each critical gate. The observability stack was deployed in a separate `monitoring` namespace and consisted of Prometheus The Prometheus Authors (2026),

Table S5 Weekly ranking utility (E15, post hoc): mean over weeks of the Spearman correlation between predicted score and realised label, and precision at the top 20%.

Horizon and metric	Formula	HANFIS	Persistence
4 weeks (45 weeks): Spearman	0.994	0.969	0.992
4 weeks: precision at top 20%	0.969	0.973	0.969
8 weeks (37 weeks): Spearman	0.989	0.963	0.982
8 weeks: precision at top 20%	0.957	0.957	0.941
12 weeks (29 weeks): Spearman	0.974	0.951	0.964
12 weeks: precision at top 20%	0.924	0.952	0.883

Table S6 Paired weekly differences in ranking utility (E15, post hoc); 95% week-bootstrap intervals in brackets.

Comparison	Δ Spearman	Δ precision at top 20%
4 w: formula − persistence	+0.002 [0.001, 0.004]	0.000 [−0.013, 0.013]
4 w: HANFIS − persistence	−0.023 [−0.027, −0.018]	+0.004 [−0.009, 0.018]
4 w: formula − HANFIS	+0.025 [0.021, 0.029]	−0.004 [−0.013, 0.000]
8 w: formula − persistence	+0.007 [0.004, 0.012]	+0.016 [0.000, 0.032]
8 w: HANFIS − persistence	−0.018 [−0.026, −0.011]	+0.016 [0.000, 0.032]
8 w: formula − HANFIS	+0.026 [0.020, 0.031]	0.000 [0.000, 0.000]
12 w: formula − persistence	+0.010 [0.005, 0.017]	+0.041 [0.014, 0.076]
12 w: HANFIS − persistence	−0.013 [−0.025, −0.002]	+0.069 [0.034, 0.103]
12 w: formula − HANFIS	+0.024 [0.017, 0.032]	−0.028 [−0.055, −0.007]

Table S7 Input-change strata at four weeks (E16, post hoc): rows, mean squared error (persistence / formula / HANFIS), and skill against persistence with 95% intervals.

Stratum	Rows	MSE	Skill, formula	Skill, HANFIS
Input changed	628	0.637 / 0.382 / 0.557	0.400 [0.308, 0.487]	0.125 [−0.402, 0.333]
No input change	452	0.107 / 0.106 / 0.202	0.002 [−0.069, 0.037]	−0.900 [−4.171, −0.206]

Prometheus Operator, kube-state-metrics, and node-exporter components. In its validated ON state, the stack contained five fully ready and running monitoring pods, including node-level exporters. In the OFF condition, the deployments and stateful workload were scaled to zero and the exporter DaemonSet was made unschedulable through a controlled node-selector patch. The original replica counts and Kubernetes object definitions were saved before the transition so that the monitoring state could be restored deterministically.

System preparation was organised as a sequence of auditable gates. The gates cover the baseline workload, cluster health, the observability deployment, the measurement procedure, protocol freezing and the generation of SHA-256 manifests; the criterion applied at each gate is listed in Table S1 of Additional file 1. The first gates of the protocol-development phase (O5–B4.10 to O5–B4.11B) are kept in the repository as scripts and describe the procedure only; since one of their gate reports could not be reconstructed after the fact, they are not cited as evidence of passing. The evidence that the gates were satisfied is the set of run archives and manifests for Runs 7–16

Table S8 Calibration of four-week out-of-fold predictions (E21, post hoc). Mean error (prediction minus label), and slope β and intercept α of the regression of the label on the prediction, with 95% image-bootstrap intervals. Unaffected rows are those whose label window does not straddle the change of the EPSS model.

Predictor	Mean error	Slope β	Intercept α
All rows (1080)			
Persistence	−0.097 [−0.196, −0.013]	0.997 [0.986, 1.007]	0.109 [0.012, 0.228]
Formula	−0.040 [−0.106, 0.017]	0.998 [0.990, 1.005]	0.048 [−0.021, 0.132]
HANFIS	0.016 [−0.080, 0.104]	1.025 [1.008, 1.044]	−0.115 [−0.228, 0.007]
Unaffected rows (890)			
Persistence	−0.015 [−0.079, 0.025]	0.997 [0.993, 1.001]	0.026 [−0.017, 0.091]
Formula	0.012 [−0.029, 0.040]	1.000 [0.997, 1.002]	−0.012 [−0.044, 0.035]
HANFIS	0.068 [−0.029, 0.157]	1.031 [1.014, 1.053]	−0.193 [−0.300, −0.085]

Table S9 E25 per fold: median over three repetitions of the training time, the peak memory increase and the single-row latency of the full flat ANFIS (F), FlatANFISMatched (M) and HANFIS (H). Unregistered descriptive measurement, first frozen seed, one machine.

	Training time (s)			Memory increase (MiB)			Latency (μs)		
Fold	F	M	H	F	M	H	F	M	H
0	0.87	0.23	5.71	0.27	5.56	0.11	92.5	47.1	395.3
1	2.17	0.57	7.66	2.48	66.75	0.25	94.1	49.1	387.6
2	11.03	0.95	9.43	4.99	194.13	0.46	96.4	50.0	402.1
3	7.30	2.71	8.42	8.20	390.17	0.85	95.6	53.8	398.2
4	12.06	6.37	10.09	6.72	663.46	0.72	97.1	47.3	424.2

Table S10 Frozen training configuration of the confirmatory comparison (amendments A-09 and A-16).

Item	Value
HANFIS membership functions	Gaussian, $m = 2$ on each raw input, $m_I = 3$ on v and e, softplus-constrained non-negative slopes (monotone)
HANFIS training	hybrid closed-form and Adam, alternating lower-panel scheme; learning rate 10^{-2}, ridge 10^{-3}
Restarts and stopping	three pre-registered restarts per fold and seed, lowest validation loss kept; patience 100, minimum 200 and maximum 600 epochs
Seeds	20260928, 20261005, 20261012, 20261019, 20261026; held-out predictions averaged over the five seeds
Split	`GroupTimeSplit`, five folds, 30-day embargo equal to the outcome window; 15% validation carve-out per fold
FlatANFISMatched	subtractive clustering with the radius searched on each training fold so that parameters match HANFIS within ±5%; patience 100, minimum 200 epochs
Flat ANFIS (full)	grid partition with $m = 2$ on seven inputs (128 rules, 1,052 parameters); patience 60

(Section 3.3), in each of which both conditions were admitted only after the gate criteria held. Because P0 stayed at two of two ready replicas with HTTP status 200 across every admitted transition, the change in observability state did not itself interrupt the target service.

Table S11 Membership functions of the documented HANFIS (fold 0, first frozen seed), in normalised input units. The last column states whether the centres keep the order of the membership-function index (one entry per input).

Panel	Input	Function	Centre c	Width s	Centres in index order
V	x_1	MF1	0.153	0.174	yes
V	x_1	MF2	0.546	0.454	
V	x_2	MF1	0.682	0.176	no
V	x_2	MF2	0.282	0.385	
V	x_3	MF1	-0.099	0.176	yes
V	x_3	MF2	0.371	0.468	
E	x_4	MF1	0.631	0.390	yes
E	x_4	MF2	0.998	0.184	
E	x_5	MF1	0.347	0.257	yes
E	x_5	MF2	1.072	0.174	
E	x_6	MF1	-0.061	0.194	yes
E	x_6	MF2	0.593	0.284	
I	v	MF1	0.352	0.379	no
I	v	MF2	0.677	0.315	
I	v	MF3	0.659	0.252	
I	e	MF1	0.317	0.244	no
I	e	MF2	0.297	0.328	
I	e	MF3	0.484	0.362	
I	x_7	MF1	0.218	0.280	yes
I	x_7	MF2	0.605	0.296	

Table S12 Rules of panels V and E of the documented HANFIS (fold 0, first frozen seed): membership-function index per input (inputs in the order x_1, x_2, x_3 for V and x_4, x_5, x_6 for E) and the consequent $f = b_0 + b_1 z_1 + b_2 z_2 + b_3 z_3$ in normalised units (slopes after the softplus constraint).

Rule	Antecedent	b_0	b_1	b_2	b_3
V1	MF1, MF1, MF1	+0.001	0.000	0.000	0.000
V2	MF1, MF1, MF2	+0.001	0.000	0.000	0.000
V3	MF1, MF2, MF1	+0.839	0.000	0.000	0.000
V4	MF1, MF2, MF2	+0.760	0.014	0.000	0.037
V5	MF2, MF1, MF1	+0.001	0.000	0.000	0.000
V6	MF2, MF1, MF2	+0.044	0.037	0.003	0.036
V7	MF2, MF2, MF1	+0.621	0.001	0.001	0.001
V8	MF2, MF2, MF2	+0.767	0.000	1.420	0.000
E1	MF1, MF1, MF1	+0.086	0.000	0.025	0.000
E2	MF1, MF1, MF2	+0.078	0.172	0.000	0.000
E3	MF1, MF2, MF1	+0.085	0.024	0.060	0.000
E4	MF1, MF2, MF2	-0.030	0.004	0.000	0.084
E5	MF2, MF1, MF1	-0.002	0.030	0.000	0.005
E6	MF2, MF1, MF2	+0.019	0.000	0.000	0.000
E7	MF2, MF2, MF1	+0.057	0.040	0.040	0.011
E8	MF2, MF2, MF2	-0.005	0.000	0.000	0.000

Supplementary note S2: Ablation and robustness analysis (original text of Section 4.5)

Table S2 of Additional file 1 reports the completed membership-family×monotonicity ablation (E1) on two synthetic scenarios designed to stress-test the training scheme before it is committed to real data. S1 is a smooth response with no thresholds and S2

Table S13 Rules of the integration panel I of the documented HANFIS (fold 0, first frozen seed): membership-function index for (v, e, x_7) and the consequent $f = b_0 + b_1 v + b_2 e + b_3 x_7$ in normalised units. A negative intercept with small slopes gives a negative output that is clipped at zero.

Rule	Antecedent (v, e, x_7)	b_0	b_1	b_2	b_3
I1	MF1, MF1, MF1	-0.529	0.000	0.000	0.004
I2	MF1, MF1, MF2	-0.002	0.008	0.000	0.084
I3	MF1, MF2, MF1	-0.665	0.000	0.000	0.005
I4	MF1, MF2, MF2	-0.002	0.005	0.000	0.109
I5	MF1, MF3, MF1	-0.455	0.000	0.000	0.004
I6	MF1, MF3, MF2	-0.002	0.005	0.000	0.072
I7	MF2, MF1, MF1	+0.136	0.758	0.074	0.000
I8	MF2, MF1, MF2	+0.027	0.030	0.008	0.000
I9	MF2, MF2, MF1	+0.080	0.808	0.080	0.000
I10	MF2, MF2, MF2	+0.022	0.020	0.010	0.006
I11	MF2, MF3, MF1	+0.092	0.613	0.061	0.000
I12	MF2, MF3, MF2	+0.019	0.020	0.007	0.000
I13	MF3, MF1, MF1	-0.457	0.217	0.000	0.005
I14	MF3, MF1, MF2	+0.030	0.037	0.005	0.103
I15	MF3, MF2, MF1	-0.623	0.166	0.000	0.007
I16	MF3, MF2, MF2	+0.032	0.035	0.007	0.135
I17	MF3, MF3, MF1	-0.406	0.158	0.000	0.004
I18	MF3, MF3, MF2	+0.023	0.028	0.005	0.089

a response containing a hard threshold; each is fit with five seeds at the 176-parameter (Gaussian) or 147-parameter (Ruspini) configuration.

The monotonicity constraint costs little on the smooth scenario but is expensive on the threshold scenario for the Gaussian family specifically (RMSE 0.117 → 6.067). A hard threshold is locally non-monotone, and softplus-constrained slopes cannot represent the local reversal. The connected Ruspini family degrades far less under the same constraint (0.646 → 2.126), because its partition of unity concentrates the approximation error near the threshold instead of distributing a large penalty across every rule. Because the real target in Section 3.4 is not known in advance to be smooth or threshold-like, amendment A-10 retains the Gaussian family with monotonicity active as the frozen default (consistent with the higher-is-riskier requirement) and treats Ruspini as a documented, cheaper fallback if the monotone-Gaussian cost proves large on real folds. An end-to-end pure-Adam variant without the closed-form anchor (Section 3.2) was also tested during this ablation and diverged on a subset of seeds under otherwise identical settings, confirming that the hybrid scheme is necessary.

Sensitivity to the EPSS model change. The EPSS scoring model changed version on 2026-06-15 (`v2025.03.14` to `v2026.06.15`). Because the label of week t is computed from snapshots $t{+}7, \ldots, t{+}28$ days ahead, two of the five held-out blocks of the confirmatory comparison involve labels computed under the newer version: one block straddles the change and one lies wholly after it. This contrast was not pre-registered, so it is reported as a post hoc sensitivity analysis. The confirmatory pipeline was re-run with the same code, seeds and frozen configuration on the 4,560 asset-weeks whose labels all pre-date the change (weeks up to 2026-05-11; 720 held-out rows), with the results in Table S3 of Additional file 1. H1a, H1c and H2 remain confirmed, and H1b remains unconfirmed, with a much wider interval. Because both the training and



Fig. S1 Methodological mind map linking the cloud-native testbed, telemetry acquisition, monitoring-overhead controls, reproducibility evidence, and HANFIS integration.

the held-out windows change, skill scores are not directly comparable across the two columns. The larger margin over the matched flat model (H2: 0.665 to 0.913) comes from the matched model, whose mean squared error per held-out row rises from 1.22 to 3.77 while that of HANFIS stays at 0.41 and 0.33.

Where HANFIS and persistence differ. The pooled skill against persistence (0.015) hides two regimes, and this exploratory, post hoc analysis (not part of the Holm-corrected family) separates them. Dropping one held-out fold at a time leaves it between 0.007 and 0.172 except for fold 3, whose removal gives -1.168. Fold 3 ends at the EPSS model change, and the label of week t is built from snapshots up to $t{+}28$ days ahead whereas the persistence forecast reuses the label of week $t{-}4$. For the four weeks in which the label is fully computed under the new model while the lagged label

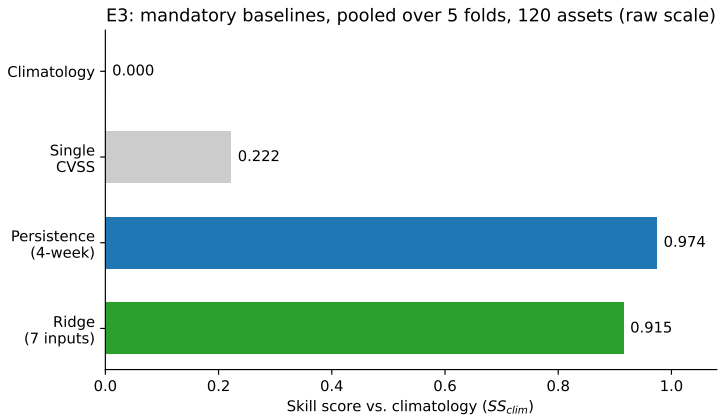


Fig. S2 Mandatory baselines' skill against climatology, pooled over 5 folds. A single CVSS score now adds modest skill over climatology (22%, up from essentially nothing on the earlier 27-image pilot); persistence and ridge separately recover over 90% of the climatology-relative skill each, though not of each other; see the mandatory-baselines table in the main text for the reversal in their relative standing.

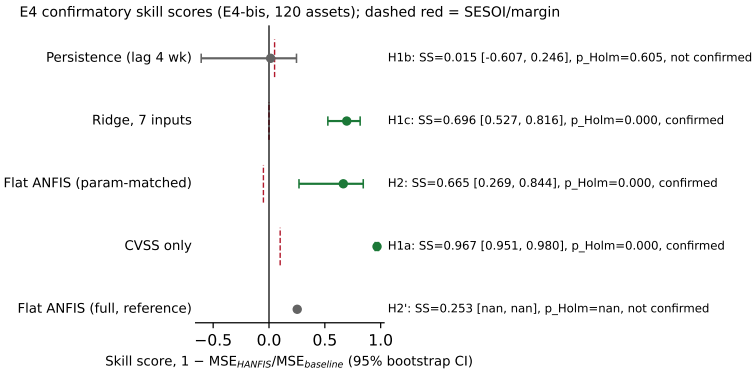


Fig. S3 E4 confirmatory results: HANFIS skill score against each comparator with 95% group-clustered bootstrap confidence intervals. Green bars exclude zero (confirmed); the red bar (H1b, vs. persistence) does not.

is still computed under the old one (2026-W24 to W27; 95 rows), skill against persistence is +0.483; on the remaining 985 rows it is −0.420. In 2026-W25 and W26 the squared error of persistence is 63.1 and 28.9 against 0.7 and 1.0 for HANFIS, while the parameter-matched model also adapts (4.8 and 5.2), so the advantage reflects access to the current EPSS input rather than the hierarchical structure. In the week the label first jumps (W24) neither HANFIS nor persistence anticipates it (108.9 and 115.2). In the stable weeks that follow HANFIS keeps a small error floor (about 0.7–0.8 per week over 24 rows, against 0.1 for persistence). The pooled advantage of 6.5 squared-error units is thus the sum of +104.1 in the break weeks and −97.6 elsewhere. The same pattern appears before the change: in the run restricted to earlier weeks the advantage

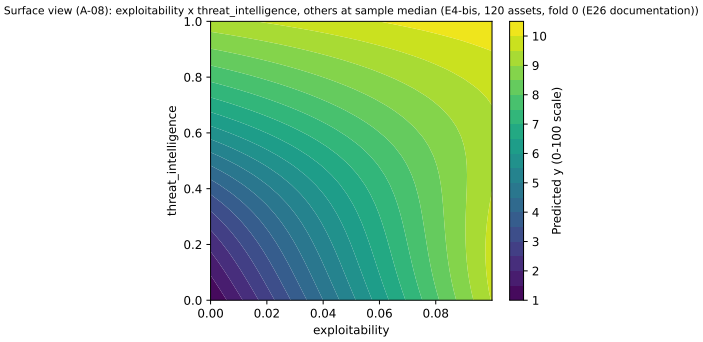


Fig. S4 Response of the five-seed ensemble of fold 0 over exploitability (x_2) and threat intelligence (x_7), the other inputs held at their sample medians (E27, descriptive, forward pass of the saved models, no training).

concentrates in 2026-W06 and W07, the weeks of one image's label surge (an image carrying 77.8% of the persistence error), and the total is −12.3. HANFIS is therefore reactive rather than predictive: better than persistence when the label moves, worse when it does not. Two further observations qualify H2. The first held-out fold, trained on 2.2 observations per parameter, contributes 91% of the pooled gap to the matched model (squared error 869.6 against 73.9). Without it the skill against FlatANFIS-Matched is 0.183 and against the full flat model 0.084, whereas the skill against the ridge model stays between 0.656 and 0.793 whichever fold is removed.

Supplementary note S3: Explanation fidelity (original text of Section 4.6)

We tested whether the rule-level explanation that HANFIS exposes is faithful to its inference. The test uses the 25 saved models (five folds × five seeds), which reproduce the confirmatory predictions of Section 4 exactly (mean over the five seeds; maximum difference 1.4×10^{-6} on the 0–100 scale in single precision). The hypotheses and thresholds were drafted and announced before execution; the version-control freeze, however, preceded the commit that contained the protocol text, so the protocol is not frozen in the strict sense and the run is reported as a documented deviation. Table S4 of Additional file 1 lists the outcome.

Removing the three rules ranked highest by attribution changed the output 4.1 times more than removing three random rules, but it exceeded the median random removal in only 83.5% of non-degenerate rows (criterion: 90%); 456 of 5,400 rows (8.4%) were degenerate, 446 of them because the output was clamped. Of the 575 positive-control pairs, 552 increased strictly; the remaining 23 include 20 pairs whose output is clamped at zero in both conditions. Post hoc and without bearing on the decision, the proportion reached 0.90 in two of five folds and was lowest in fold 3 (0.637), and it was lowest on the P0 and P0-IF rows (0.648, ratio 1.73, 135 rows). On the nine paired test weeks the mean ON–OFF contrast was 0.037 for the model and 0.073 for the label (0–100 scale): the direction is right and the magnitude is about half.

We therefore do not claim validated rule-level explanations. The rule trace matches the inference exactly and the direction of the response is verified, but the attribution of the output to individual rules is faithful on average rather than uniformly.

Supplementary note S4: Prioritisation utility (original text of Section 4.12)

Risk scores are used to rank assets, so we also compared the predictors on ranking within each week rather than on squared error. For every week with at least 15 assets we computed the Spearman correlation between the predicted score and the realised label and the precision at the top 20% (the share of the 20% of assets with the highest predicted score that are also among the 20% with the highest realised label), and averaged over weeks. Differences between predictors are paired by week, with 95% intervals from a bootstrap that resamples weeks, a more conservative unit than images. This analysis is post hoc and uses the out-of-fold predictions at the three horizons.

All predictors rank assets almost identically (Table S5 of Additional file 1). At four weeks the weekly Spearman correlation is 0.99 for the formula and for persistence and 0.97 for HANFIS, and the precision at the top 20% is 0.97 for all three, so the lower squared error of the formula does not translate into a practical gain in prioritisation (Table S6 of Additional file 1: Spearman +0.002 and precision 0.000 against persistence). At eight and twelve weeks persistence ranks somewhat worse (precision 0.94 and 0.88) and both HANFIS and the formula improve on it in precision (+0.016 and +0.041 for the formula, +0.016 and +0.069 for HANFIS). At twelve weeks HANFIS identifies the top 20% better than the formula (+0.028, interval [0.007, 0.055]) although its Spearman correlation is lower (−0.024). Differences in Spearman correlation never exceed 0.03 and the precision is close to its ceiling, so all these differences should be read as small; with eighteen paired comparisons and no correction for multiplicity, the intervals are descriptive.

Supplementary note S5: Error anatomy and the EPSS model change (original text of Section 4.13)

The preceding sections compare pooled errors. To see where the differences come from, we analysed the four-week out-of-fold predictions by stratum, by time block and by distance to the change of the EPSS model on 2026-06-15 (post hoc, with a bootstrap clustered by image).

Reactivity. The advantage of the formula over persistence is concentrated in rows in which an input changed during the previous four weeks (EPSS exploitability or KEV status; 628 of 1,080 rows). Its skill is 0.400 (interval 0.308 to 0.487) in these rows and 0.002 (−0.069 to 0.037) where nothing changed, so it behaves like persistence when nothing happens (Table S7 of Additional file 1); a gate that returns persistence in unchanged rows leaves it unchanged. HANFIS has skill 0.125 in changed rows and −0.900 (−4.171 to −0.206) in unchanged rows: it adds noise where persistence is already adequate. A gate removes this deficit (improvement 0.098, interval 0.033 to 0.204) but leaves HANFIS no better than persistence (0.111, −0.367 to 0.292) and

worse than the formula (−0.383, −0.998 to −0.131). Since the flag counts any change of the score, most flagged changes are small.

Squared error is extremely concentrated: the worst 5% of rows hold 91% (persistence), 95% (formula) and 67% (HANFIS) of the total, and the five worst images 68%, 63% and 39%. Leaving out each of the five time blocks in turn leaves the skill of the formula against persistence between 0.34 and 0.43, with intervals above zero, although one block (April to June 2026) carries 73% of the absolute reduction in error; HANFIS is worse than the formula in every such estimate.

The model change. The largest errors are not KEV events (the KEV indicator is zero in all twelve worst rows of every predictor) but small shifts of the score, with labels between 3.7 and 10 points on a 100-point scale, in the weeks 2026-W21 to W24. The four-week label windows of these weeks straddle the change of the EPSS model: x_2 is computed with the old model while the label, and the label of the persistence forecast, use the new one, and all three predictors err almost equally there. Rows within the label window of the change (190 of 1,080 at four weeks) hold 77% of the squared error of persistence and of the formula and 49% of that of HANFIS. On the remaining rows (Table S16) the formula keeps its point estimate of skill against persistence (0.348 against 0.357) but the interval now includes zero (−0.060 to 0.422). HANFIS is significantly worse than persistence (−1.228) and than the formula (−2.418). The forward evaluation described above is free of this change: its test weeks start on 2026-08-31 and its persistence labels come from weeks after 2026-08-03.

Horizon, revisited. The narrowing of the gap between HANFIS and the formula at eight and twelve weeks (Section 4.11) must be read in this light. Excluding the affected rows, HANFIS stays behind the formula at eight and twelve weeks (−0.324 and −0.362, intervals including zero). On the 169 asset-weeks that are common to the three horizons and unaffected at all of them, the skill of HANFIS against the formula still rises with the horizon (−15.4, −1.1 and −0.5; change between four and twelve weeks +14.9, interval 6.1 to 53.7). It stays significantly negative at every horizon (−0.515, interval −2.117 to −0.092, at twelve weeks). The horizon sweep therefore shows a shrinking deficit, not parity; the sample of unaffected rows is small and the intervals are wide.

Supplementary note S6: Calibration, recalibration and stacking, E21 to E24 (original text of Section 4.14)

We checked calibration after the registration of the forward evaluation had become public (post hoc, descriptive, no training). For the four-week out-of-fold predictions of persistence, the formula and HANFIS we report the mean error (prediction minus label), the slope β and intercept α of the regression of the label on the prediction (a calibrated predictor has $\beta = 1$ and $\alpha = 0$), and the mean label by decile of the prediction. Intervals come from the bootstrap clustered by image used above. The coefficients of the formula were fitted inside each training fold, on the same folds as HANFIS, so its calibration is also out-of-fold. Table S8 of Additional file 1 covers all 1,080 rows and the 890 rows whose label window does not straddle the change of the EPSS model (Section 4.13).

All three predictors lie close to the identity line. Persistence runs low on all rows (mean error −0.097, interval −0.196 to −0.013), and almost all of this bias comes from the weeks around the model change: on unaffected rows it is −0.015 (−0.079 to 0.025). The interval of the formula's mean error includes zero in both samples. HANFIS departs most: its slope is 1.025 (1.008 to 1.044) on all rows and 1.031 (1.014 to 1.053) on unaffected rows, where its intercept is −0.193 (−0.300 to −0.085). In the fourth decile of its predictions the mean prediction is 1.0 and the mean label 0.5 (unaffected rows: 1.1 and 0.5); the mean label does not rise between the third and fourth decile (0.5 in both) although the mean prediction doubles. In the upper deciles predictions and labels agree (7.9 and 8.1 in the eighth decile, 9.8 and 9.9 in the ninth). These departures are small next to the label scale. The intervals above resample images and treat the weekly shocks shared by all images as fixed. A bootstrap clustered by week (E22, post hoc; 45 weeks, 37 of them unaffected by the model change) gives narrower intervals for these statistics, not wider ones: for HANFIS the slope is 1.025 (1.015 to 1.035) on all rows and 1.031 (1.020 to 1.042) on unaffected rows, and the intercept −0.115 (−0.189 to −0.037) and −0.193 (−0.242 to −0.147). To compare predictors we took the distance from the ideal value ($\beta = 1$, $\alpha = 0$, mean error 0) and subtracted that of the formula from that of HANFIS, so that a positive number means HANFIS is further away. For the slope the excess is 0.023 (0.011 to 0.033) on all rows and 0.031 (0.018 to 0.042) on unaffected rows; with intervals clustered by image the first is 0.023 (0.001 to 0.041), a borderline result. On all rows the excess in the intercept and in the mean error includes zero (0.068, −0.072 to 0.179, and −0.024, −0.061 to 0.061; week-clustered). On unaffected rows the excess in the slope and in the intercept is positive under both kinds of clustering (slope 0.031: week 0.018 to 0.042, image 0.013 to 0.051; intercept 0.182: week 0.131 to 0.230, image 0.063 to 0.278). The excess in the mean error (0.055), by contrast, is positive only with the week-clustered interval (0.034 to 0.068; image: −0.016 to 0.133). Against persistence the pattern on unaffected rows is the same: slope 0.028 (week 0.015 to 0.040, image 0.011 to 0.050) and intercept 0.167 (week 0.111 to 0.219, image 0.021 to 0.286) are positive under both, the mean error 0.053 (week 0.005 to 0.088, image −0.053 to 0.146) only with week clustering. We read the miscalibration of HANFIS as small but consistent in slope and intercept where the model change does not interfere, and as descriptive.

Recalibration (E23). We asked whether a linear correction could close the gap between HANFIS and the formula. For this we refitted label $= a + b\times$ prediction for each predictor, once on the rows of the other folds (cross-fitted) and once on all rows (in-sample, an optimistic ceiling), and recomputed the skill of HANFIS against the formula (post hoc; intervals clustered by image). Cross-fitting uses rows from later folds, so this is a diagnostic and not a forecasting procedure. On all rows the skill is −0.534 (−1.383 to −0.204) before, −0.535 (−1.287 to −0.232) after the cross-fitted and −0.509 (−1.223 to −0.213) after the in-sample recalibration; the change from the cross-fitted recalibration is −0.001 (−0.047 to 0.116). On unaffected rows the values are −2.418 (−13.204 to −0.709), −2.315 (−12.609 to −0.702) and −2.175 (−12.388 to −0.647); the change of 0.103 (−0.152 to 1.060) includes zero and is about 4% of the deficit (10% for the in-sample ceiling). The deficit of HANFIS is therefore not

explained by a linear miscalibration: it remains after the correction, in line with the noise HANFIS adds in rows where no input changed (Table S7 of Additional file 1).

Stacking (E24). We asked whether HANFIS carries information that the formula lacks. For this we fitted label $= a + b_1 \times$ formula $+ b_2 \times$ HANFIS on the rows of the other folds, applied it to each fold in turn (cross-fitted), and compared it with the formula recalibrated in the same way (post hoc; intervals clustered by image). On all rows the skill of the combination against the recalibrated formula is -0.001 (-0.006 to 0.004); the best case, with coefficients fitted on all rows, is 0.0005 (-0.002 to 0.003), and the weight of HANFIS is 0.030 (between folds 0.015 to 0.052). Persistence as the second predictor instead of HANFIS gives 0.001 (-0.022 to 0.021). On unaffected rows the values are -0.002 (-0.028 to 0.017) for HANFIS and -0.169 (-0.224 to -0.005) for persistence; the second reflects unstable coefficients, since the weights of formula and persistence vary widely between folds when both are included. A linear combination of the two predictors therefore does not beat the formula: HANFIS adds no information that a straight line can use.

Supplementary note S7: Limitations (original text of Section 5, subsection Limitations)

Two limits of construct validity come first. First, the label and two of the inputs share ingredients: x_2 (EPSS $\times\, \rho$) and x_7 (the KEV indicator) are the values at t of quantities that the label aggregates over $t{+}7, \ldots, t{+}28$, and runtime reachability ρ enters both. The task is therefore to forecast the near future of exploitability signals that the model can already see, not to predict independent compromise outcomes, for which no ground truth exists here. The persistence baseline, built in the same way, is the relevant bar, and the comparison with a single CVSS score should be read in that light. The exact Shapley attribution (Section 4.7) and the ablation without these two inputs (Section 4.8) show how heavily the model's skill rests on them. Second, the dynamic character of the score comes from the time-varying EPSS and KEV feeds and from runtime reachability. Runtime telemetry (CPU, memory, latency) is used only to quantify observability overhead and is not a model input, so "dynamic" must not be read as telemetry-driven scoring. The rule-level attribution is faithful on average but not uniformly (Section 4.6), and we do not claim validated explanations.

All analyses labelled post hoc were designed after the confirmatory results and run on the same data; they are descriptive, carry no multiplicity correction, and none of them changes a registered decision.

The inventory consists of public container images and a small number of testbed assets built from a public base image, and the label reflects exploitation signals of public vulnerabilities (EPSS, KEV), not incidents in a specific organisation; generalisation to other workloads or to incident-based outcomes is untested.

The confirmatory comparison (Table 3) uses the full 120-asset inventory. On the same inventory (6,357 labelled rows; Table 2), a linear ridge model over all seven inputs underperforms persistence on every one of the five grouped-temporal folds ($\mathrm{SS}_{\mathrm{pers}} = -2.293$ pooled, raw scale). We suspect that the linear model over-extrapolates as the inventory's heterogeneity grows, where persistence's week-to-week copy remains comparatively safe, but we did not test this explanation.

The EPSS scoring model changed version (`v2025.03.14` to `v2026.06.15`) inside the sealed 57-week window, a threat to label comparability across that boundary. Two of the five held-out blocks involve labels computed under the newer version, and a post hoc re-analysis restricted to weeks whose labels all pre-date the change leaves the confirmatory conclusions unchanged (Section 4.5).

The data-sufficiency gate (at least ten observations per parameter) is met by the pooled sample (6,357 labelled rows against 176 parameters) but not by the two earliest held-out folds (2.2 and 7.1 observations per parameter), which are also those in which HANFIS is furthest behind persistence. H2's pooled margin is dominated by the first fold (Section 4.5).

CVEs lacking a Trivy-reported publication date are dated by their first appearance in a sealed EPSS snapshot, which is conservative (at most 7 days late) but not exact. A single CVE whose Trivy status lies outside the accepted set (for example "under investigation" for a freshly published Red Hat advisory) makes the whole asset-week unusable. This affected one of the 120 images for seven consecutive weeks from 2026-08-10, and it can reach at most the two Red Hat based images of the inventory. Such rows are removed from the analysis.

Labels whose four-week window straddles the EPSS model change of 2026-06-15 (and the persistence labels derived from them) mix two score definitions; these rows (18% of the out-of-fold rows) hold 77% of the squared error of persistence and of the formula, and the forward evaluation avoids them.

The label aggregation rule (a row-wise maximum over active CVEs), its numeric scale, and the operational definition of Patch Risk were closed and pre-registered as amendment A-13 (cross-checked against the literal target-definition formula, Section 3.4) before the confirmatory comparison in Table 3 was run.

The testbed is a single two-node WSL 2/Docker Desktop Kubernetes cluster; external validity to production multi-node clusters and other CNI/storage stacks is untested. Ten paired observability runs, while showing large and consistent effects, still provide limited power for null-hypothesis testing on the latency measure specifically, which is why the reported confidence intervals rather than a significance threshold carry the interpretive weight in Section 4.1.

Supplementary material moved from the main text on 11 November 2026

The following material stood in the main text until 11 November 2026 and was moved here so that the main text reports only the headline numbers. Nothing was re-run; the numbers are those of the original sections.

Supplementary note S8: Observability-overhead protocol (external-v3.0) and validity safeguards (text removed from Section 3.3 on 11 November 2026)

The gates and integrity controls of Table S1 of Additional file 1 were developed during the first runs and were carried forward unchanged into protocol `external-v3.0`,

Table S14 Paired monitoring overhead over ten counterbalanced runs (protocol external-v3.0). CI: 10,000-resample bootstrap; p: exact two-sided sign-flip test (floor $2/2^{10} = 0.002$).

Measure	Mean Δ	95% CI	Relative	p
Resource cost of the observability stack				
Node CPU (mCPU)	+99.36	[96.29, 103.35]	+43.3%	0.002
Monitoring namespace CPU (mCPU)	+39.79	[33.00, 52.77]	—	0.002
Node memory (MiB)	+280.76	[250.99, 303.54]	+18.7%	0.002
Monitoring namespace memory (MiB)	+204.19	[191.38, 218.82]	—	0.002
Effect on the observed workload				
Latency median (ms)	−0.005	[−0.055, +0.044]	−0.06%	0.859
Latency P95 (ms)	+0.059	[−0.126, +0.265]	+0.67%	0.607
Latency P99 (ms)	+0.188	[−0.127, +0.448]	+1.94%	0.268
P0 container CPU (mCPU)	−0.059	[−0.156, +0.041]	−3.9%	0.305
P0 container memory (MiB)	−0.334	[−0.522, −0.147]	−1.19%	0.016

which produced all observability-overhead results reported in Section 4.1. The protocol comprises ten complete counterbalanced ON–OFF pairs (Runs 7–16): Runs 8, 10, 12, 14 and 16 were executed OFF–ON, and Runs 7, 9, 11, 13 and 15 ON–OFF. The earlier Runs 1–6 belong to the protocol-development phase and are not pooled with the ten pairs. One aborted attempt of Run 7 and one aborted attempt of Run 17 failed the integrity gate and are recorded as incidents rather than observations.

The protocol follows a fail-closed rule. A condition is admitted to measurement only when the nodes, the P0 workload (two of two replicas), the HTTP service (status 200) and the required monitoring state satisfy the acceptance criteria of Table S1 of Additional file 1. A failed gate yields a `HOLD` decision, not an observation. Each run is archived with a SHA-256 manifest, so that the ten run archives and the cumulative analysis can be re-verified without reconstructing the original command sequence.

Telemetry preserves the link between each value and its Kubernetes object, node, container, condition and run. The measurements are node-, pod- and container-level CPU and memory, pod readiness, HTTP outcome and response latency; identical request-generation and sampling procedures are applied in the ON and OFF conditions. Configuration files, event logs, gate reports and manifests document how each observation was produced and are not treated as HANFIS training features.

Let $Y_{r,c,k}$ denote observation k of run r under condition $c \in \{\mathrm{ON}, \mathrm{OFF}\}$, where Y may be CPU, memory or latency. The run mean is

$$\bar{Y}_{r,c} = \frac{1}{n_{r,c}} \sum_{k=1}^{n_{r,c}} Y_{r,c,k}, \tag{1}$$

the paired absolute overhead is $\Delta Y_r = \bar{Y}_{r,\mathrm{ON}} - \bar{Y}_{r,\mathrm{OFF}}$, and the relative overhead, when the OFF denominator is non-zero, is

$$\Delta Y_r^{(\%)} = \frac{\bar{Y}_{r,\mathrm{ON}} - \bar{Y}_{r,\mathrm{OFF}}}{\bar{Y}_{r,\mathrm{OFF}}} \times 100\%. \tag{2}$$

The run, not the individual sample, is the independent replication unit, which avoids pseudoreplication from temporally adjacent samples. Inference over the ten paired differences uses an exact two-sided sign-flip test, whose smallest attainable p with ten pairs is $2/2^{10} = 0.002$, together with a 10,000-resample bootstrap Efron (1979) confidence interval (fixed seed) for the mean difference. Effect sizes, confidence intervals and the consistency of direction carry more interpretive weight than a binary significance decision. The estimated increment is an instrumentation cost; it must not be read as a change in application exposure.

Validity safeguards and methodological boundary.

Internal validity rests on paired comparisons, a stable workload, condition gates, health checks, counterbalanced order, and the rejection of measurements taken under invalid states. Construct validity is protected by separating observability overhead from cybersecurity exposure and by requiring a semantic mapping before any telemetry variable is used as a HANFIS feature. Reliability is supported by script-based execution, frozen configuration and cryptographic integrity checks, and traceability runs from raw samples through summaries to model-ready features.

Several limits remain. WSL 2 with Docker Desktop is reproducible but does not represent every production Kubernetes deployment, and the two-node topology and P0 workload constrain external validity. Across the ten pairs of Runs 7–16 (external-v3.0), the two nodes together used on average about 230 mCPU without and 329 mCPU with the monitoring stack (node-level CPU summed over both nodes). That is 1.4% and 2.1% of the host's 16 logical cores; the capacity was taken from the v4.2 collector metadata and `docker info` because it was not logged during these runs. The absence of a latency effect was therefore observed at these utilisation levels and need not extend to saturated nodes. Ten pairs make an effect-size reading appropriate, but they bound the smallest attainable sign-flip p at 0.002 and give limited power for small latency effects. The protocol was frozen before the ten pairs were analysed, yet its version is entangled with the platform on which the earlier runs were taken, so cross-platform comparisons are descriptive. Finally, successful telemetry collection does not by itself show predictive value for cyber risk; that claim rests on the held-out HANFIS evaluation against the baselines of Section 4.

Supplementary note S9: Horizon sweep and feature impact by horizon (full text of Section 4.11)

All preceding analyses use the four-week label window of the confirmatory protocol. To check whether the conclusions depend on that horizon, we rebuilt the dataset with label windows of 58 and 86 days (eight and twelve weekly snapshots after t). To avoid leakage we set the persistence lag and the embargo to the window length (8 and 12 weeks; 58 and 86 days). We then repeated the out-of-fold comparison of HANFIS, the max-form formula and persistence with the same folds, seeds and hyperparameters. Longer windows leave fewer labelled weeks (the last labelled week is 2026-07-27 and 2026-06-29), so the out-of-fold samples shrink to 889 and 697 rows, and mean squared errors at different horizons are not comparable because the labels average different numbers of snapshots; only skill scores are compared. This analysis is post hoc.

The formula's advantage over persistence is stable (skill 0.36, 0.49 and 0.44 at four, eight and twelve weeks), whereas the gap between HANFIS and the formula closes: −0.534 (interval [−1.438, −0.208]) at four weeks, −0.048 ([−0.205, 0.032]) at eight and 0.039 ([−0.031, 0.110]) at twelve. HANFIS beats persistence from eight weeks on (0.467 and 0.462, intervals above zero). Because the longer horizons only contain older weeks (the EPSS model changed on 2026-06-15), horizon and period are confounded. Restricting all three horizons to the 408 asset-weeks present in each, the pattern remains: the skill of HANFIS against the formula changes by +0.862 ([+0.309, +2.973]) between four and twelve weeks, while the skill of the formula against persistence does not change (−0.016, [−0.096, +0.141]).

Removing exploitability and the KEV indicator again removes all skill at eight and twelve weeks (skill against CVSS alone −0.099 and −0.179, intervals including or touching zero; against HANFIS with all inputs −24.0 and −14.6). A ridge model on the remaining five inputs is not better than CVSS alone (−0.069 and −0.071). The relative gain of HANFIS at longer horizons does not come from the other inputs; Section 4.13 shows that it is largely due to weeks around the EPSS model change and that on unaffected rows HANFIS remains behind the formula. The evidence is suggestive, not conclusive: at twelve weeks HANFIS and the formula are not distinguishable, cross-fitted blends of the two do not improve on either alone, and the forward evaluation of Section 4.16 covers only the four-week horizon.

Supplementary note S10: Error anatomy and the EPSS model change (full text of Section 4.13)

The preceding sections compare pooled errors. To see where the differences come from, we analysed the four-week out-of-fold predictions by stratum, by time block and by distance to the change of the EPSS model on 2026-06-15 (post hoc, with a bootstrap clustered by image).

Reactivity. The advantage of the formula over persistence is concentrated in rows in which an input changed during the previous four weeks (628 of 1,080 rows): its skill is 0.400 (interval 0.308 to 0.487) there and 0.002 (−0.069 to 0.037) where nothing changed (Table S7 of Additional file 1). HANFIS has skill 0.125 in changed rows and −0.900 (−4.171 to −0.206) in unchanged rows: it adds noise where persistence is already adequate. A gate that returns persistence in unchanged rows removes this deficit but leaves HANFIS no better than persistence (0.111, −0.367 to 0.292) and worse than the formula (−0.383, −0.998 to −0.131).

The model change. The largest errors are small shifts of the score in the weeks 2026-W21 to W24, whose four-week label windows straddle the EPSS model change: x_2 is computed with the old model while the label, and the label of the persistence forecast, use the new one. Rows within the label window of the change (190 of 1,080 at four weeks) hold 77% of the squared error of persistence and of the formula and 49% of that of HANFIS. On the remaining rows (Table S16) the formula keeps its point estimate of skill against persistence (0.348 against 0.357) but the interval now includes zero (−0.060 to 0.422); HANFIS is significantly worse than persistence (−1.228) and than the formula (−2.418). The forward evaluation (Section 4.16) is free of this change. At eight and twelve weeks the gap between HANFIS and the formula

Table S15 Horizon sweep (E14, post hoc). Skill score with 95% image-clustered bootstrap interval. Panel A: all out-of-fold rows; B: the 408 asset-weeks present at all three horizons; C: HANFIS retrained without x_2 and x_7 against CVSS only.

Comparison	Skill	95% interval	Rows
A. All out-of-fold rows			
4 weeks: formula vs. persistence	0.357	[0.279, 0.407]	1080
4 weeks: HANFIS vs. persistence	0.015	[−0.620, 0.243]	1080
4 weeks: HANFIS vs. formula	−0.534	[−1.438, −0.208]	1080
8 weeks: formula vs. persistence	0.492	[0.269, 0.696]	889
8 weeks: HANFIS vs. persistence	0.467	[0.236, 0.651]	889
8 weeks: HANFIS vs. formula	−0.048	[−0.205, 0.032]	889
12 weeks: formula vs. persistence	0.440	[0.245, 0.597]	697
12 weeks: HANFIS vs. persistence	0.462	[0.277, 0.610]	697
12 weeks: HANFIS vs. formula	0.039	[−0.031, 0.110]	697
B. Same 408 asset-weeks at every horizon			
4 weeks: HANFIS vs. formula	−0.812	[−2.952, −0.264]	408
8 weeks: HANFIS vs. formula	−0.045	[−0.218, 0.038]	408
12 weeks: HANFIS vs. formula	0.050	[−0.060, 0.158]	408
Change 12 minus 4 weeks: HANFIS vs. formula	0.862	[0.309, 2.973]	408
Change 12 minus 4 weeks: formula vs. persistence	−0.016	[−0.096, 0.141]	408
C. HANFIS without x_2, x_7 vs. CVSS only			
4 weeks	−0.135	[−0.336, 0.046]	1080
8 weeks	−0.099	[−0.289, 0.069]	889
12 weeks	−0.179	[−0.406, 0.007]	697

narrows (Section 4.11), but on rows unaffected by the change HANFIS stays behind the formula (−0.324 and −0.362, intervals including zero): the horizon sweep shows a shrinking deficit, not parity. The concentration of error, the leave-one-block-out estimates and the common-weeks analysis are in Supplementary note S5 of Additional file 1.

Table S16 Skill on all out-of-fold rows and on rows unaffected by the EPSS model change (E19, post hoc), with 95% image-bootstrap intervals from a separate run (they differ slightly from Table S15). Unaffected rows: 890 of 1,080 (four weeks), 536 of 889 (eight) and 342 of 697 (twelve).

Comparison	All rows	Unaffected rows
4 w: formula vs. persistence	0.357 [0.279, 0.407]	0.348 [−0.060, 0.422]
4 w: HANFIS vs. persistence	0.015 [−0.627, 0.252]	−1.228 [−12.804, −0.015]
4 w: HANFIS vs. formula	−0.534 [−1.383, −0.204]	−2.418 [−13.204, −0.709]
8 w: formula vs. persistence	0.492 [0.269, 0.696]	0.623 [−0.009, 0.699]
8 w: HANFIS vs. persistence	0.467 [0.241, 0.670]	0.501 [−1.631, 0.702]
8 w: HANFIS vs. formula	−0.048 [−0.200, 0.036]	−0.324 [−2.153, 0.052]
12 w: formula vs. persistence	0.440 [0.245, 0.597]	0.824 [0.270, 0.913]
12 w: HANFIS vs. persistence	0.462 [0.269, 0.614]	0.761 [−0.220, 0.888]
12 w: HANFIS vs. formula	0.039 [−0.034, 0.106]	−0.362 [−1.342, 0.018]

Supplementary note S11: Accuracy-complexity curve (full text of Section 4.4)

The scalar inter-tier bottleneck between the lower panels and the integration panel is a candidate explanation for any residual gap between HANFIS and a full-capacity flat ANFIS: compressing v and e to a single number may discard cross-subsystem interaction that the flat model can still represent. A sweep over the bottleneck width k, widening v and e from scalars to k-dimensional vectors passed into the integration panel, traces this trade-off directly. Grid-partitioning the integration tier, however, makes its rule count grow as $m_I^{2k} \cdot m$. At the frozen $m_I = 3$, $k = 1$ reproduces HANFIS exactly (34 rules, 176 parameters), $k = 2$ already costs 178 rules and 1,152 parameters (6.5x), and $k \geq 3$ exceeds 1,400 rules (Figure 2), which is more parameters than the present inventory has rows. The sweep script refuses these settings by default rather than attempting an underdetermined fit. We ran $k \in \{1, 2\}$; $k = 4$, originally planned, was dropped on these grounds. This combinatorial explosion is itself read as evidence for why a scalar bottleneck was the design choice in the first place, independent of what $k = 1$ vs. $k = 2$ shows empirically.

At k=1 the sweep uses exactly A-09's training scheme: the alternating closed-form least squares for both lower panels, the intercept refit after the monotone projection, and the same hard clamp on v, e that HANFIS's own forward pass applies. It is implemented separately in `hanfis_bottleneck.py` under a multi-head representation that generalises cleanly to $k > 1$. A per-row comparison with the confirmatory predictions on the earlier 109-asset inventory confirmed this reproduction directly (Pearson $r = 0.9984$ on point predictions, identical fold assignment for all 981 held-out rows); the comparison was not repeated on the 120-asset inventory. An earlier version of this sweep used a simpler training scheme (the Integration-only closed form rather than the alternating scheme) and omitted the intercept refit after the monotone projection; both have since been corrected, and the numbers below supersede that earlier, confounded comparison.

k=1 recovers skill broadly consistent with HANFIS's own H1b result (Table 3: $\mathrm{SS}_{\mathrm{pers}} = 0.015$, $[-0.607, 0.246]$). Both confidence intervals cross zero and overlap substantially, as the per-row parity above predicts, though the point estimates (0.169 vs. 0.015) agree less closely than on the smaller inventory. k=2's interval now excludes zero entirely on the negative side. On the completed 120-asset inventory, widening the bottleneck is significantly worse than persistence. On the smaller inventory it had been unsupported by the evidence but not shown to be worse. The paired comparison on matched resampled images (the same resample is used for both arms, so sampling noise is shared) gives $\mathrm{SS}_{\mathrm{pers}}(k{=}1) - \mathrm{SS}_{\mathrm{pers}}(k{=}2) = 0.791$, 95% CI $[0.354, 2.037]$, which excludes zero (100% of resamples favour k=1). k=1 is also pooled-RMSE-better on all five folds (two-sided sign test $p = 0.0625$, the floor attainable with five folds). This is the same margin of evidence as on the smaller inventory, even though each arm's marginal significance against persistence flipped in opposite directions.

This is the strongest statement the present inventory supports. It is a mechanism claim, not a claim that wider bottlenecks are intrinsically worse: every k=2 fold falls under ten observations per parameter (range 0.3–2.9), the same ten-per-parameter threshold applied elsewhere in this study (Section 3.4), while k=1 clears it in three of

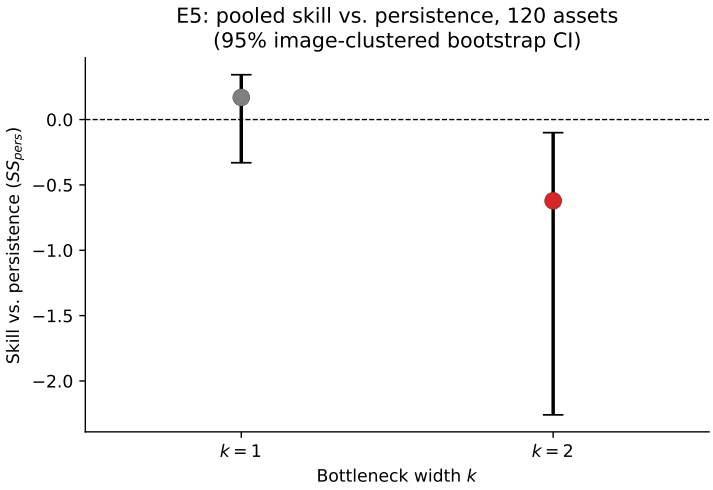


Fig. S5 E5: pooled skill against persistence for $k \in \{1, 2\}$ with 95% image-clustered bootstrap confidence intervals. k=1's interval now crosses zero; k=2's is entirely negative.

five folds. A secondary diagnostic separates part of this gap from pure data scarcity. Disabling the hard clamp on v, e (an ablation, not the model HANFIS reports) costs k=1 0.161 of skill but costs k=2 0.813. Both were re-run on the completed 120-asset inventory; the costs were 0.019 and 0.394 respectively on the 27-image pilot. This is consistent with v, e being estimated less reliably from the same training rows once six times as many consequent parameters compete for them, so that excursions outside $[0, 1]$ are both more frequent and more consequential once clamped. Separating an intrinsic cost of wider bottlenecks from this data-scale effect requires an inventory substantially larger than the 120 assets collected here, which is left for future work. The architecture's own combinatorial growth in k (Figure 2) means that this separation gets harder, not easier, as k increases.

References

Efron B (1979) Bootstrap methods: Another look at the jackknife. The Annals of Statistics 7(1):1–26. https://doi.org/10.1214/aos/1176344552

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Table S17 Map from the experiment and amendment codes used in the project repository to the sections of the main text an

Code	Status
E1	pre-registered, synthetic
E3	pre-registered
E4, E4-bis	pre-registered (tag e4-prereg-2026-09-29); E4-bis is the same protocol on the completed 120-asset inven
E5	pre-registered sweep
E6	documented deviation
E7	pre-registered, OSF-registered forward evaluation
E8	post hoc
E9 to E11	post hoc
E14	post hoc
E15	post hoc
E16 to E19	post hoc
E21 to E24	post hoc
E25	unregistered, descriptive
E26	post hoc
E27	descriptive documentation
Runs 7–16	protocol external-v3.0
A-02	frozen setting
A-04	frozen setting
A-06	frozen setting
A-09	frozen setting
A-10	frozen setting
A-13	frozen setting
A-16	frozen setting